

Attacks on Vision-Language-Action Robotic Systems: A Systematic Evidence Review

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Abstract—Vision-Language-Action (VLA) models turn perception and language into executable robot behavior. This creates an attack surface that is not reducible to adversarial image classification, text-only jailbreaking, conventional robot robustness, or broad VLA safety: a small visual, linguistic, training-time, or system-level intervention can propagate through grounding, planning, action representation, controller execution, and recovery. This survey reviews how VLA attacks emerge in closed-loop robotic systems, how the literature has evolved from adversarial vision, prompt injection, and robot-policy attacks toward direct VLA patches, black-box instruction attacks, action-level backdoors, trajectory manipulation, reasoning-chain hijacking, and privacy attacks, and what current evidence can and cannot support. We use a structured evidence review to separate direct VLA attacks from embodied-adjacent evidence and broader contextual literature, while keeping the main narrative focused on attack mechanisms and robotic consequences. Because attack success is not yet used consistently in VLA safety work, we formalize typed VLA attack success as a rollout-level violation predicate tied to attacker assumptions, benign baselines, validation tier, and consequence. We introduce an orthogonal attack map spanning lifecycle, entry surface, target component, target representation, security property, knowledge level, access channel, privilege level, validation tier, and consequence. Across studies, the evidence shows rapid growth but also clear limits: direct VLA results remain concentrated around OpenVLA/LIBERO-style settings, physical validation is uneven, and controller, recovery, and pre/post-controller traces are rarely reported. We close with a robotics-oriented evaluation protocol and a research roadmap for physically grounded, action-representation-aware, and recovery-aware VLA attack studies.

Index Terms—Vision-Language-Action Attacks, Robot Foundation Models, Systematic Evidence Review, Embodied Security, Physical Attack Evaluation.

I. INTRODUCTION AND POSITIONING

Vision-Language-Action (VLA) models and robot foundation models are moving robot autonomy from task-specific perception–planning–control stacks toward policies that jointly interpret images, language, robot state, and interaction history before producing actions. PaLM-E, RT-1, RT-2, VIMA, Open X-Embodiment, OpenVLA, Octo, and π_0 illustrate this shift toward web-scale multimodal pretraining, large robot datasets, cross-embodiment transfer, and natural-language task specification [1]–[8]. Related systems such as SayCan, Code as Policies, ProgPrompt, VoxPoser, CLIPort, Diffusion Policy, and ACT show that language and vision can also supervise skill selection, code generation, waypoint synthesis, trajectory distributions, and action-chunk prediction [9]–[15]. For robotics, this is not only a new model family. It is a new control interface whose inputs are semantically rich and whose outputs can move robots, operate tools, write memory, and change the future observation stream.

This interface makes attacks qualitatively different from ordinary VLM, LLM, or robot-policy failures. In a VLM, an adversarial image may yield a wrong answer; in a text-only LLM, a jailbreak may yield unsafe text; in a conventional robot policy, a disturbance may reduce control performance. In a VLA-enabled robot, these effects can meet inside a closed-loop embodied process. A printed patch can change visual grounding, a small instruction edit can redirect a trajectory, a scene-text injection can cross a source boundary, a poisoned demonstration can implant a trigger, and an action-level backdoor can force a gripper primitive at a critical moment. The consequence may appear as a wrong object, unauthorized tool use, action freezing, contact risk, privacy leakage, or persistent state compromise rather than as a single incorrect label.

A. Related Surveys and Positioning

The closest adjacent work is the VLA safety survey *Vision-Language-Action Safety: Threats, Challenges, Evaluations, and Mechanisms* [16]. That survey covers the broader safety landscape: threats, alignment, evaluation, defense mechanisms, and deployment challenges. Our paper does not try to recover that landscape with a different emphasis. It asks a different set of review questions: *Which attacks have actually been demonstrated against VLA models or VLA-controlled robots? What evidence level supports each claim? Which target representation was changed? How was attack success defined? Which results are direct VLA evidence, which are embodied-adjacent, and which are only analogical?*

This distinction matters because the literature is now moving quickly from position statements to concrete VLA attack papers. Direct evidence includes adversarial prompt and patch attacks on VLA policies [17]–[20], action freezing and action/trajectory backdoors [21]–[26], transferable and model-agnostic patch attacks [27], [28], 3D object-texture attacks [29], CoT-reasoning hijacking [30], trajectory redirection [31], and VLA membership inference [32], [33]. These papers are not just examples inside a safety taxonomy; they define the empirical base that an authoritative attack survey must compare.

B. Research Questions

The review is organized around five research questions.

- **RQ1:** Which attacks have been validated directly on VLA models, reasoning-enabled VLAs, VLA-controlled robots, or VLA training/privacy channels?

TABLE I
INCREMENT OVER ADJACENT SURVEYS. THE COMPARISON IS ABOUT REVIEW ARTIFACTS, NOT THE VALUE OF EACH LITERATURE.

Dimension	Closest VLA safety survey	Embodied security survey	LLM-agent security survey	This review
Direct VLA attack corpus	Discusses VLA threats within broader safety.	Usually sparse or absent.	Usually absent.	Freezes 23 direct VLA attack records and separates them from 7 core adjacent attacks and 55 contextual references.
Evidence grading	Threat/evaluation oriented.	Vulnerability oriented.	Software-agent risk oriented.	Direct, embodied-adjacent, contextual; core/context corpus split; highest validation tier and validation flags.
Action representation	Often covered as model capability.	Usually not central.	Not applicable.	Codes action token, chunk, waypoint, trajectory, velocity/flow, skill/API, controller, and memory representations.
Controller/recovery	Defense or safety mechanism discussion.	Cyber-physical component discussion.	Not applicable.	Reports coverage gaps for chunk horizon, controller frequency, safety filter, recovery policy, and pre/post-controller traces.
Privacy attacks	Usually safety-adjacent.	Sometimes privacy/security.	Often data leakage.	Treats VLA membership inference and VLALeaks as direct VLA confidentiality evidence.
Reusable data product	No attack-specific per-paper coding database.	No VLA attack evidence database.	Benchmarks but not robot VLA coding.	Provides frozen CSV database, corpus/source file, direct audit file, validation flags file, and README.
Quantitative findings	Broad trend synthesis.	Broad vulnerability synthesis.	Benchmark scores or attack rates.	Reports counts and proportions for direct VLA evidence, validation, model/benchmark concentration, and reporting completeness; does not pool incompatible ASR.

- **RQ2:** Which lifecycle stages, entry surfaces, target components, target representations, and action representations do these attacks affect?
- **RQ3:** What knowledge level, access channel, and privilege level do current attacks assume?
- **RQ4:** What validation tier do current studies reach, and do they report real actuation, controller behavior, safety filters, or recovery?
- **RQ5:** Where do current evaluations remain weak in comparability, transfer, physical validation, and robotics safety interpretation?

Table I compares this review against the closest adjacent survey traditions. The difference is not simply that we are “attack-centered.” We ask how attack mechanisms interact with VLA action representations, closed-loop execution, controller boundaries, and the strength of the available evidence.

C. Scope and Definition

We treat a *VLA attack* as an intentional intervention that changes a VLA-enabled system’s perception, grounding, planning, policy decoding, execution, memory, tool use, privacy boundary, or human-robot interaction in a way that violates the intended task, authorization boundary, privacy expectation, or safety envelope. This definition separates adversarial attacks from ordinary task failure and non-adversarial robustness failure. A robot that fails because of unmodeled friction exhibits robustness risk; a robot that fails because an adversary selected a perturbation, instruction, tool response, memory entry, trigger, or physical scene configuration to induce a violation exhibits security risk.

We include three evidence strata. *Direct evidence* attacks VLA models or VLA-controlled robots. *Embodied-adjacent*

evidence attacks embodied LLM/VLM planners, robot policies, middleware, or HRI systems that share action or physical consequence boundaries but are not end-to-end VLA policies. *Analogical evidence* comes from adversarial classifiers, autonomous sensing, software agents, RAG, general RL, or robot cybersecurity when it informs a VLA attack surface but does not by itself establish a VLA result. We exclude ordinary distribution shift without an adversarial objective, purely text-only jailbreak work without an embodied implication, and image-classification attacks that do not help analyze grounding, planning, action, privacy, or physical consequence.

D. Contributions

This paper discusses attacks at a responsible academic level and does not provide operational payloads, exploit recipes, or instructions for real-world misuse. It makes five contributions.

- We reposition the paper as a systematic evidence review of attacks on VLA-enabled robotic systems, with explicit contrast to VLA safety and embodied-AI surveys.
- We provide a transparent evidence base that separates direct VLA attacks, embodied-adjacent attacks, and contextual references without letting background work inflate attack claims.
- We formalize typed VLA attack success, separating task failure, targeted behavior, safety violation, authorization violation, privacy leakage, persistence, and recovery failure.
- We replace mixed one-dimensional categories with an orthogonal attack map covering lifecycle, entry surface, target component, target representation, security property, attacker assumptions, validation tier, and consequence.
- We synthesize direct VLA attacks through their mechanisms and robotic consequences, including cross-study

findings on target models, validation, action representations, controller reporting, and recovery gaps.

- We propose a standardized evaluation protocol for VLA attacks that connects typed ASR, benign utility, transfer, query budget, physical validation, closed-loop recovery, and consequence severity.

II. REVIEW PROTOCOL AND EVIDENCE BASE

We use an adapted PRISMA-style reporting workflow; the protocol was not prospectively registered. The goal is to make the survey’s evidence base transparent without letting the protocol dominate the story. We distinguish a *Core Evidence Corpus*, containing attack studies with an explicit attacker, target, method, and outcome, from a *Contextual Reference Corpus*, containing model substrates, benchmarks, adversarial-ML methods, robot-security work, and action-representation foundations. The frozen supplementary files provide the full record list, field dictionary, audit trail, and commands for regenerating the counts used in the main text.

A. Databases, Dates, and Search Strings

The last search for this revision was conducted on 16 June 2026. The search time range was all records indexed or publicly available by that date; no lower date bound was imposed, but direct VLA attack screening emphasized 2023–2026. We searched Google Scholar, Semantic Scholar, arXiv, IEEE Xplore, ACM Digital Library, USENIX proceedings, OpenReview, CVF Open Access, ACL Anthology, PMLR, RSS/CoRL/ICRA/IROS proceedings pages, and project/repository pages linked from papers. The supplementary README lists source-specific query templates, discovery-source codes, and commands for regenerating the statistics.

The core search strings were instantiated with spelling variants and venue filters:

- ("vision-language-action" OR VLA OR "robot foundation model") AND (attack OR adversarial OR jailbreak OR backdoor OR poisoning OR privacy OR "membership inference")
- ("OpenVLA" OR "RT-2" OR "Octo" OR "pi0" OR "LIBERO") AND (patch OR prompt OR backdoor OR "action token" OR "action chunk" OR trajectory)
- ("embodied AI" OR "LLM-controlled robot" OR "robotic agent") AND ("prompt injection" OR jailbreak OR red-team OR attack)
- ("robot" OR ROS OR "cyber-physical") AND (security OR sensor attack OR middleware attack OR HRI attack)
- ("adversarial patch" OR "physical adversarial" OR LiDAR OR RAG OR "tool agent") AND (robot OR embodied OR VLA)

Targeted verification searches were then run for direct VLA attack names and classes mentioned in recent papers: DropVLA/TabVLA, VLA-Fool, SABER, Q-DIG, BadVLA,

GoBA, AttackVLA/BackdoorVLA, FreezeVLA, EDPA, AD-VLA, UPA-RFAS, VLA-Hijack, Tex3D, TRAP, ReasonBreak, SilentDrift, trajectory redirection, partially observable patch attacks, and membership inference/VLALeaks.

Backward snowballing followed references from seed VLA attack papers, VLA/RFM surveys, the closest VLA safety survey, and embodied-security surveys when a cited work either attacked a relevant system or defined an attacked substrate. Forward snowballing used citation search from seed direct VLA attack papers and was stopped when new results no longer met inclusion criteria or were duplicates. Targeted verification search was used only after a named attack or project was encountered in a paper, review, project page, or reviewer suggestion; this reduces name-missing errors but is treated as a possible bias in Section VII.

B. Inclusion and Exclusion Criteria

We included a work in the *Core Evidence Corpus* if it satisfied at least one of the first two criteria: (i) it directly attacks a VLA model, reasoning-enabled VLA, VLA-controlled robot, VLA training channel, or VLA privacy channel; or (ii) it attacks an embodied LLM/VLM planner, robot policy, HRI channel, physical sensing path, or embodied agent with an explicit attacker and action/robot consequence. We included a work in the *Contextual Reference Corpus* if it provided attack-evaluation methodology, a foundation model, benchmark, action representation, robot middleware/security substrate, or adversarial-ML mechanism necessary to interpret the core evidence.

We excluded works that only report ordinary robustness or distribution shift without an adversarial objective; text-only jailbreaks with no embodied, tool, or action boundary; image-classification attacks with no grounding, planning, action, or physical consequence relevance; broad AI safety position papers without attack or evaluation mechanisms; and records whose bibliographic identity could not be verified from an arXiv, publisher, OpenReview, project, or repository page. Duplicate records were merged when they referred to the same paper across arXiv, project page, repository, OpenReview, or conference page. When both preprint and venue versions existed, the peer-reviewed venue version was used as the bibliographic record and the arXiv/project page was retained only for version or artifact notes.

C. Evidence Levels

Every included record is labeled by corpus type and evidence stratum. The *Core Evidence Corpus* contains direct VLA attacks and embodied-adjacent attacks with explicit attacker, target, method, and outcome. The *Contextual Reference Corpus* contains analogical and methodological records used to define substrates, mechanisms, benchmarks, or evaluation concepts. *Direct evidence* means the victim is a VLA model or a VLA-controlled robot system. *Embodied-adjacent evidence* means the victim is an embodied LLM/VLM planner, a traditional robot policy, a physical sensing path, or an HRI system with physical or action consequences but not a VLA policy. *Contextual/analogical evidence* provides mechanisms

TABLE II
TRANSPARENT SCREENING FLOW FOR FROZEN DATABASE
VLA-ATTACK-ER-V2026-06-16. COUNTS DESCRIBE THIS
MANUSCRIPT’S ADAPTED PRISMA-STYLE REPORTING WORKFLOW; THE
PROTOCOL WAS NOT PROSPECTIVELY REGISTERED.

Screening stage	Count	Notes
Records captured before de-duplication	276	Database queries, arXiv/OpenReview/CVF pages, seed surveys, backward/forward snowballing.
Duplicate or superseded records removed	51	Same work across arXiv, project page, repository, conference version, or renamed preprint.
Title/abstract screened	225	Screened for VLA, embodied robotics, adversarial objective, privacy objective, or evaluation relevance.
Excluded at title/abstract	113	Ordinary robustness only, no embodied implication, text-only safety, or unrelated robotics.
Full texts assessed	112	Full paper, arXiv page, publisher page, OpenReview page, project page, or repository was checked.
Full texts excluded	27	No adversarial objective (8); no action/robot boundary (6); insufficient VLA relevance (7); unverifiable identity (4); short duplicate (2).
Core Evidence Corpus	30	Direct VLA attacks (23) plus embodied-adjacent attacks with explicit attacker, target, method, and outcome (7).
Contextual Reference Corpus	55	VLA/RFM substrates, benchmarks, adversarial-ML methods, RAG/security-agent context, ROS/cyber-physical security, and action-representation foundations.
Included records in frozen database	85	Core evidence (30) plus contextual references (55).

or evaluation lessons from classifiers, autonomous-driving perception, RAG, software agents, general RL, ROS, or cyber-physical systems. Cross-study findings about attacks use the core corpus as the denominator unless a direct-VLA subset is explicitly named.

We used the following decision rule for direct evidence. A record is coded as *direct VLA* only if (1) the target is a named VLA model, a reasoning-enabled VLA, a VLA-controlled robot, or a VLA training/privacy channel; and (2) the measured outcome concerns VLA actions, trajectories, action representations, reasoning-to-action behavior, triggered VLA behavior, or VLA training-data membership. If the target is an embodied LLM/VLM planner, robot middleware, HRI channel, or sensor stack without a VLA policy measurement, the record is *embodied-adjacent*. If the work provides an attack mechanism, benchmark substrate, or representation baseline without embodied VLA validation, it is *analogical/methodological*. Ambiguous records are conservatively downgraded rather than upgraded.

D. Coding Reliability and Audit Procedure

The 23 direct VLA records were independently re-coded in a second audit pass for principal fields: evidence stratum, family, lifecycle, knowledge level, access channel, privilege level, highest validation tier, target representation, objective, action representation, real-robot actuation, transfer reporting, query-budget reporting, and recovery reporting. The second coder was given the frozen CSV and coding rules but not the manuscript’s narrative claims. Initial agreement was 23/23 for evidence stratum, family, lifecycle, knowledge/access coding, highest validation tier, target representation, objective, and action representation. Initial agreement for real-robot actuation was 21/23; reconciliation changed RoboGCG and VLALeaks from ambiguous/reported actuation to no real-robot actuation, while FreezeVLA was kept conservative as NR. Reporting counts were regenerated from the reconciled CSV.

We also audited a stratified non-direct sample of 18 records: 10 embodied-adjacent and 8 contextual/analogical. Agreement was 17/18 for strict core/context placement and 18/18 for lifecycle, access channel, highest validation tier, target representation, and objective under metadata-only review. The one flagged case was software-agent evidence, which is retained as contextual unless it has robot or embodied action consequence. We report percent agreement rather than Cohen’s κ because the audited sample is small and several fields are multi-label with many NR/NA values. For multi-label fields, disagreements were resolved by conservative downgrading or NR. The direct-audit file and README are supplied with the database supplement.

E. Supplementary Evidence Files

The supplement contains five files: the per-paper evidence table, a corpus/source table, a direct-VLA audit table, a direct-VLA validation-flags table, and a README with field dictionary and commands for regenerating the main statistics. The most important rule is orthogonality: the same paper can have a vision entry surface, an action-token target representation, an integrity objective, black-box knowledge level, physical access channel, simulator validation, and task-failure consequence. Those are different axes, not competing categories.

The structured evidence base is used in three ways. First, it prevents literature inflation: direct VLA attacks, embodied-adjacent attacks, and contextual references are not treated as the same kind of evidence. Second, it prevents metric inflation: ASR values are not pooled when they refer to different objectives, denominators, validation settings, or robot consequences. Third, it makes gaps visible: a paper can be strong direct evidence for simulator action deviation and still weak evidence for physical transfer, recovery, or adaptive attacks against runtime monitors.

III. VLA SYSTEM, THREAT MODEL, AND ORTHOGONAL TAXONOMY

A. Closed-Loop VLA System Model

A VLA-enabled robot can be viewed as a closed-loop state transition system rather than a single input-output model. Let

x_t denote the physical state, $o_t = O(x_t, \eta_t)$ the observation produced by sensors and preprocessing noise η_t , m_t the memory/tool/human context, and u_t the command emitted by the VLA or its planner. A low-level controller K maps u_t and robot state to actuation, while a recovery or safety mechanism R may stop, replan, filter, or ask for help:

$$\begin{aligned} u_t &= \pi_\theta(o_{\leq t}, g, m_t), \\ a_t &= K(u_t, x_t, R_t), \\ x_{t+1} &= F(x_t, a_t, w_t), \quad m_{t+1} = U(m_t, o_t, u_t, a_t). \end{aligned} \quad (1)$$

The safety envelope is a set $\mathcal{S}$ over physical state, controller state, tool authorization, privacy state, and human proximity. An attack is relevant when an intervention δ changes one of these transitions or contexts so that the rollout violates a task, authorization, privacy, or safety predicate. This formulation keeps the useful parts of formalism: it identifies the observation function, VLA policy, controller, memory update, recovery mechanism, and safety constraint set that a paper must report.

We use the propagation chain

$$\begin{aligned} \delta &\rightarrow \Delta o_t \rightarrow \Delta z_t \rightarrow \Delta u_t \rightarrow \Delta a_t \\ &\rightarrow \Delta x_{t+1} \rightarrow \text{recovery or violation} \end{aligned} \quad (2)$$

as the organizing model for robotics evidence. The same entry perturbation can stop at the observation, change an embedding or grounding representation, change a VLA command, be clipped or transformed by a controller, or only become visible as a later state violation. RQ4 and RQ5 are therefore evaluated at the controller and recovery boundary, not only at the model-output boundary.

B. Typed Attack Success

Because VLA safety papers do not yet use a uniform definition of attack success, we define it explicitly. Let $\tau^0(c)$ be the benign rollout from an initial condition and task context $c = (x_0, g, m_0)$, and let $\tau^\delta(c)$ be the rollout under an adversarial intervention δ that satisfies the declared knowledge, access, privilege, budget, and physical constraints. A VLA attack is successful for objective type O and consequence class C when

$$\begin{aligned} \text{Succ}_{O,C}(\tau^\delta, \tau^0) &= \mathbf{1}\{\phi_{O,C}(\tau^\delta) = 1 \\ &\quad \wedge \phi_{O,C}(\tau^0) = 0 \wedge \delta \in \Delta(\mathcal{A})\}. \end{aligned} \quad (3)$$

where $\phi_{O,C}$ is a typed violation predicate and $\Delta(\mathcal{A})$ is the admissible attacker set. The benign rollout condition prevents ordinary task difficulty or non-adversarial distribution shift from being counted as attack success. If a benign rollout is unavailable, the paper should state the baseline used, such as clean task success, human-labeled intended behavior, or non-member trajectory statistics.

The predicate $\phi_{O,C}$ must be typed. A *task-failure attack* succeeds when the benign task would complete but the attacked rollout fails. A *targeted behavior attack* succeeds when the robot reaches an attacker-specified object, action primitive, waypoint, trajectory, skill, or tool call. A *safety attack* succeeds when the rollout violates a safety envelope,

contact margin, force/velocity/workspace bound, or human-proximity constraint. An *authorization attack* succeeds when the system executes an unauthorized command, skill, tool, or privilege transition. A *privacy attack* succeeds when membership, trajectory, user, or scene information is inferred above a stated threshold. A *persistence attack* succeeds when the adversarial effect survives replanning, feedback, memory update, fine-tuning, or a session boundary. A *recovery-failure attack* succeeds when the system has an opportunity to stop, replan, ask for help, or invoke a recovery policy but still reaches the violation predicate.

This definition makes ASR meaningful only when five items are stated together: the violation predicate, the denominator, the attacker assumptions, the validation tier, and the benign or clean baseline. Without these items, ASR values from different VLA papers should be treated as local evidence rather than comparable field-level rates.

C. Threat Model

We assume the deployer has an intended task, authorized users, authorized tools and skills, and a safety envelope for people, property, data, and robot hardware. The attacker seeks to violate one or more of these constraints while preserving enough surface plausibility that the system continues acting, logs the event as normal, or stores corrupted context for future sessions.

Attacker capability is decomposed into three fields. *Knowledge level* records what the attacker knows about weights, gradients, architecture, prompts, action heads, or controllers: white-box, gray-box, black-box, transfer-only, or unknown. *Access channel* records how the intervention is delivered: digital, physical, human-mediated, supply-chain, or hybrid. *Privilege level* records the role or boundary involved: user, bystander, operator, data contributor, model provider, tool/API, middleware, or no privilege. This prevents a common coding error: treating “black-box” and “physical” as alternatives. A physical patch can be optimized with white-box gradients, transferred in a black-box setting, or deployed by a bystander with no digital privileges.

D. Orthogonal Taxonomy

The taxonomy in Table III is multi-label and orthogonal. It deliberately avoids placing perception, goal hijacking, HRI, and persistence in the same list. Perception is often an entry surface; goal hijacking is usually an integrity or authorization violation; HRI is an entry surface, access channel, or privilege setting; persistence is a temporal profile; real-robot actuation is a validation tier; collision risk is a consequence. Mixing them hides what was actually demonstrated.

The supplementary coding-decision table records boundary cases such as VLM planners with controllers, autonomous-driving VLA trajectory models, membership inference, ROS exploits, and vision-encoder-only attacks. In the main text we apply the same conservative rule: without VLA policy, VLA trajectory, VLA action representation, or VLA training/privacy measurement, the record is contextual or embodied-adjacent rather than direct.

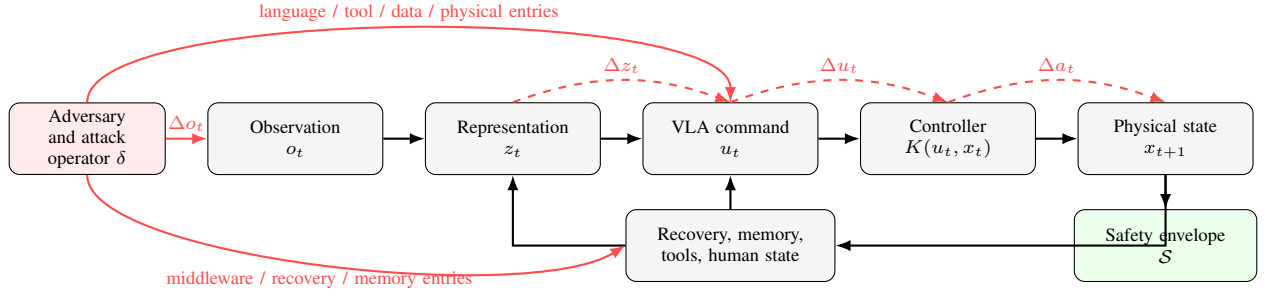


Fig. 1. Closed-loop VLA attack view. The entry surface, target representation, and consequence can occur at different points in the loop.

TABLE III
ORTHOGONAL MULTI-AXIS TAXONOMY FOR CODING VLA ATTACKS. EACH PAPER MAY RECEIVE MULTIPLE LABELS ON EACH AXIS; AXES ARE NOT MUTUALLY EXCLUSIVE CATEGORIES.

Axis	Allowed labels	Coding question
Lifecycle	Training; inference; runtime; post-deployment	When is the adversarial intervention introduced or activated?
Entry surface	Vision; language; state; memory; tool; middleware; human; environment	Which semantic or system channel carries the adversarial content?
Target component	Visual encoder; language encoder; multimodal projector; planner/reasoner; action head; controller; memory; tool/API; middleware	Which system component is attacked or measured?
Target representation	Embedding; grounding; plan; action token; chunk; waypoint; trajectory; velocity field; controller state; memory entry	Which representation or control object changes inside or across components?
Security property violated	Availability; integrity; confidentiality; authorization	Which security property is violated? Physical safety is represented as a consequence rather than merged into this axis.
Knowledge level	White-box; gray-box; black-box; transfer-only; unknown	What does the attacker know about weights, gradients, architecture, prompts, action heads, or controllers?
Access channel	Digital; physical; human-mediated; supply-chain; hybrid	How does the attacker deliver the intervention? This is separate from knowledge level.
Privilege level	User; bystander; operator; data contributor; model provider; tool/API; middleware; no privilege	What role or privilege boundary does the attacker occupy or abuse?
Temporal profile	One-shot; repeated; delayed; persistent	Does the attack act once, recur over feedback, activate later, or survive sessions/training?
Validation tier	Digital input; real sensor; simulation; HIL; robot actuation; human-proximate	What is the strongest demonstrated validation tier? Do not infer actuation or human proximity from input-only tests.
Consequence	Task failure; wrong object; unsafe action; collision/contact risk; privacy leakage; unauthorized skill; persistent state compromise	What happened at the robot/system boundary? Keep consequence separate from objective and entry surface.

E. Coding Examples

The same paper receives different labels on different axes. The ICCV patch attack on VLA robotics enters through *vision*, targets action/trajectory representations, violates integrity, and has possible safety consequences through changed robot motion [18]. SABER enters through *language*, targets plans and action sequences through black-box rollout feedback, has a repeated inference-time profile, and validates in LIBERO simulation across multiple VLA models [20]. DropVLA is a training/fine-tuning-time supply-chain attack: its entry surface is poisoned vision/text data, its target representation is an action primitive within chunked control, its security property is integrity, its consequence can be unsafe or wrong-object action, its temporal profile is delayed/persistent, and its evidence includes simulation plus a physical Franka demonstration [24]. VLA membership inference is different again: the entry surface is model query or internal signals, the target representation is training-set membership reflected in action likelihoods or temporal motion patterns, the security property is confidentiality, and the consequence is privacy harm rather than task failure [32], [33].

This coding also prevents overclaiming. A VLM jail-break with no robot action is analogical evidence for source confusion, not direct evidence of VLA unsafe action. A ROS2 middleware exploit is embodied-adjacent evidence for tool/middleware compromise, not a VLA policy attack unless the VLA system consumes or emits the compromised message. A physical LiDAR or camera attack can be strong sensor evidence but remains analogical for VLA action unless a VLA-controlled robot rollout is measured.

IV. EVIDENCE SYNTHESIS

This section synthesizes the Core Evidence Corpus defined in Section II, using contextual references only to interpret mechanisms and substrates. The main pattern is clear: direct VLA attack evidence is now substantial enough to review, but it remains concentrated in OpenVLA/LIBERO-style manipulation, often preprint-heavy, and uneven in physical validation. Embodied-adjacent attacks remain useful for threat modeling, while contextual references should not be written as if they were validated VLA attacks.

TABLE IV

EVIDENCE-AWARE SYNTHESIS BY CORPUS. CORE ROWS ARE ATTACK EVIDENCE; CONTEXTUAL ROWS SUPPORT TAXONOMY OR EVALUATION BUT ARE NOT INCLUDED IN ATTACK-STUDY DENOMINATORS. D: DIGITAL; S: SIMULATION; HIL: HARDWARE-IN-THE-LOOP; P: PHYSICAL ROBOT/WORLD; PR: PEER-REVIEWED; PP: PREPRINT OR NON-ARCHIVAL VERSION.

Corpus / stratum	Representative work	Status	Validation	What the evidence supports
Direct VLA: instruction/control	RoboGCG, VLA-Fool, SABER, Q-DIG, trajectory redirection [17], [19], [20], [31], [34]	PP	D/S/P	Language or multimodal perturbations can alter VLA rollout behavior; black-box, natural-prompt, and command-preserving settings are emerging.
Direct patches/physical vision	VLA: ICCV patch attacks, EDPA, ADVLA, UPA-RFAS, VLA-Hijack, Tex3D, partially observable patches [18], [27]–[29], [35]–[37]	PR/PP	D/S/P	Visual attacks can target embeddings, grounding, visual proprioception, action trajectories, and object textures; transfer and physical evidence are improving but uneven.
Direct action/backdoor	VLA: FreezeVLA, BadVLA, GoBA, DropVLA/TabVLA, AttackVLA/BackdoorVLA, SilentDrift [21]–[26]	PP	S/P	VLA attacks can target action primitives, action freezing, chunk drift, and long-horizon triggered behavior, often while preserving clean utility.
Direct VLA: reasoning/privacy	TRAP, altered reasoning traces, ReasonBreak, VLA membership inference, VLALeaks [30], [32], [33], [38], [39]	PP	D/S/P	Reasoning traces and training membership are VLA-specific target representations; objectives include integrity and confidentiality rather than only task failure.
Core: embodied-adjacent attacks	BadRobot, RoboPAIR, CHAI, RoboJailBench, ANNIE, Phantom Menace, Eva-VLA [40]–[46]	PR/PP	D/S/P	Embodied LLM/VLM, environmental command, trajectory, sensor, and physical-variation attacks inform VLA threat models but are not counted as direct VLA evidence unless VLA policies are attacked.
Contextual references	VLA/RFM substrates, benchmarks, adversarial ML, physical/sensor attacks, RAG/tool-agent security, RL backdoors, ROS/robot cybersecurity [6], [8], [47]–[54]	PR/PP	D/S/P	Provides mechanisms, substrates, and evaluation lessons; not included in attack-study denominators unless the work has explicit attacker, target, method, and outcome.

A. Training-Time and Supply-Chain Attacks

Direct evidence. VLA backdoor work has expanded from generic task disruption to action and trajectory control. BadVLA studies objective-decoupled VLA backdoors that preserve clean performance while inducing triggered action deviations [22]. Its evidence is direct because the victim is a VLA policy and the measured output is action behavior, but its strongest claims are still tied to the particular trigger, benchmark, and fine-tuning setup. GoBA changes the threat model by using ordinary physical objects as goal-oriented triggers and by introducing BadLIBERO-style evaluation with “nothing to do,” “try to do,” and “success to do” levels [23]. This is stronger than a binary failure label because it distinguishes a triggered attempt from completed attacker-specified behavior.

DropVLA/TabVLA narrows the target further to reusable low-level action primitives, such as gripper release, under small poisoning budgets and reports simulation plus physical feasibility [24]. Its main contribution for a robotics audience is temporal precision: the attack is not only that a task fails, but that a primitive fires at a decision window. AttackVLA then provides a broader evaluation harness for adversarial and backdoor attacks and introduces BackdoorVLA for attacker-specified long-horizon action sequences [25]. SilentDrift is especially robotics-relevant because it attacks action chunking and delta-pose integration, showing how small poisoned drifts can accumulate during intra-chunk open-loop execution [26]. Together, these papers show a progression from *triggered failure to triggered goal behavior, primitive-level control, and chunk-level trajectory drift*. They should not be collapsed into one backdoor category.

Evidence strength and gaps. The direct VLA backdoor evidence is stronger than analogies from classifiers because it measures action traces and clean-task retention. It is still not yet field-wide evidence. Most studies use curated manipulation suites and controlled triggers; few report how data inspection,

trajectory validation, controller clipping, or post-training updates affect attack survival. Few compare whether the same trigger works across autoregressive token heads, continuous action heads, action chunks, and flow/diffusion policies. These gaps matter because training-time attacks become deployment risks only if the trigger persists through the actual adaptation pipeline, robot morphology, camera placement, and controller.

Embodied-adjacent and analogical evidence. Classic poisoning and backdoor attacks establish the general supply-chain risk [55]–[58], and RL backdoors show that sequential policies can learn trigger-conditioned behavior [53], [59]–[61]. These works justify VLA backdoor threat models but do not by themselves prove VLA vulnerability. The direct VLA backdoor papers now provide that evidence, with the caveat that most remain recent preprints and frequently rely on a small set of benchmarks and robot embodiments.

B. Perception, Multimodal, and Physical Attacks

Direct evidence. Early VLA patch work formulates action-discrepancy, position-aware, and targeted manipulation objectives and evaluates digital plus physical patch settings [18]. This work is important because the objective is defined in action space rather than as image classification error. VLA-Fool expands the attack surface to textual, visual, and cross-modal misalignment attacks under white-box and black-box settings [19]. Its strongest value is not a single failure number, but the comparison across language perturbations, visual patches/noise, and image–instruction mismatch.

EDPA reduces the required attacker knowledge by targeting visual encoder embeddings and cross-modal alignment rather than the full action head [27]. ADVLA pushes a related gray-box direction by using attention-guided sparse feature-space perturbations that are low-amplitude and patch-wise localized [35]. Universal transferable patch work, such as UPA-RFAS, studies black-box transfer across VLA models, prompts, viewpoints, and sim-to-real settings [28]. VLA-Hijack targets visual

proprioception by suppressing real-arm features and injecting a phantom embodiment patch, exposing a representation-level reason for cross-architecture transfer [36]. Tex3D moves from 2D patches to adversarial object textures optimized over long-horizon trajectories and evaluated in simulation and real-robot settings [29]. Partially observable patch attacks further constrain the adversary to a short trajectory prefix, a more realistic physical-access setting [37].

These papers form a useful sequence of evidence. Action-space patches ask whether visual perturbations can change robot motion. Embedding and attention attacks ask whether attacking the shared VLM/VLA representation is enough when the action head is unknown. Transferable patches ask whether the attack survives architecture, fine-tuning, and viewpoint changes. 3D texture attacks ask whether the attack can be bound to manipulated objects rather than pasted into the camera frame. Partial-observation attacks ask whether the adversary can attack without seeing the full future rollout. The sequence is more informative than a single “visual attack” bucket.

Analogical evidence. Foundational adversarial examples, physical patches, object-detector attacks, 3D mesh attacks, LiDAR attacks, and audio-command attacks show that physical sensor inputs can be manipulated [48]–[50], [62]–[81]. Their direct implication for VLA systems depends on whether the perturbation changes grounding, plan, action, trajectory, or recovery. This is why VLA patch papers should report action traces and consequence labels rather than only image-level success.

C. Instruction, Planning, and Reasoning Attacks

Direct evidence. RoboGCG adapts adversarial prompt optimization to VLA control authority and shows that a prompt-level intervention at rollout start can influence action reachability over time [17]. Its claim is stronger than a text jailbreak claim because the target is robot action reachability, but the access and optimization assumptions still need to be reported whenever the result is cited. SABER makes the instruction threat model more practical by using a black-box, agentic edit process with bounded character/token/prompt edits and rollout-level feedback across multiple VLA models [20]. Q-DIG approaches red-teaming differently: rather than optimizing a single adversarial suffix, it uses quality-diversity search to generate natural, task-relevant instructions that expose robot-policy failures and can be used for robustness improvement [34]. Trajectory-level redirection studies command-preserving prompts that keep the instruction close to the intended task while redirecting the physical outcome [31].

Reasoning-enabled VLAs create an additional internal target. TRAP shows that CoT or intermediate reasoning traces can be hijacked by physical patches to steer downstream actions without changing the user’s instruction [30]. Altered-thought studies further indicate that entity references in VLA reasoning traces can causally affect robot manipulation success, while non-reasoning control models may be immune to that specific channel [38]. ReasonBreak extends the reasoning-trajectory question to autonomous-driving-style VLA systems

and evaluates black-box textual corruptions against reasoning and trajectory safety metrics in closed-loop simulation [39]. The shared lesson is that reasoning text is neither merely explanation nor merely prompt context; in some VLA designs it is a control-relevant representation.

Embodied-adjacent evidence. BadRobot, RoboPAIR, CHAI, SafeAgentBench, EARBench, and RoboJailBench evaluate embodied jailbreak, physical command hijacking, and risk-aware planning where success is measured closer to action than text-only refusal [40]–[43], [82], [83]. These are important for goal hijacking and HRI threat models, but they should be labeled as embodied-adjacent when the target is not a VLA policy.

D. Action-Level and Closed-Loop Attacks

Action representation is the robotics core of this review. Autoregressive action-token policies expose token probabilities and discretized gripper/pose bins; a prompt or patch can change which token is selected, and early token errors can propagate through subsequent actions. RT-1, RT-2, OpenVLA, and SpatialVLA illustrate tokenized or discretized action interfaces whose action vocabulary, spatial grids, or detokenizers become attack-relevant representations [2], [3], [6], [84]. Parallel action chunks create a different risk: the model commits to several future actions before new observations can correct the error. ACT, OpenVLA-OFT, and related high-throughput VLA fine-tuning recipes show why chunk length, temporal ensemble, and parallel decoding should be reported as security variables rather than only efficiency variables [15], [85]. SilentDrift directly exploits this intra-chunk open-loop property [26].

Waypoint and trajectory policies expose geometric objectives such as end-effector displacement, curvature, unsafe workspace entry, or targeted path redirection [18], [31]. Planner-mediated systems such as CLIPort, Code as Policies, ProgPrompt, and VoxPoser make object binding, generated code, value maps, and waypoint fields explicit enough to inspect but also explicit enough to attack [10]–[13]. Diffusion and flow policies, including Diffusion Policy and π_0 , shift the target from a token to a conditional trajectory distribution, denoising path, velocity field, or sampling process [8], [14]. Generalist policy and dataset work such as Octo, Open X-Embodiment, DROID, BridgeData, RoboNet, and Gato further matter because they define the cross-embodiment and data-provenance assumptions under which attacks may or may not transfer [5], [7], [86]–[89].

The closed-loop consequence depends on controller and recovery design. A temporal ensemble can damp high-frequency perturbations but may also prolong a poisoned action chunk. A high re-planning frequency can recover from one-frame perception attacks but may be vulnerable to repeated low-amplitude nudges. Controller latency and stale observations can turn a semantically correct command into an unsafe physical action. Safety envelopes and low-level controllers can prevent direct collision while still allowing wrong-object manipulation, unauthorized skill use, privacy leakage, or near-boundary behavior. Therefore, direct VLA attack papers

TABLE V

DESCRIPTIVE STATISTICS FROM FROZEN DATABASE VLA-ATTACK-ER-V2026-06-16. ATTACK-STUDY STATISTICS USE THE CORE EVIDENCE CORPUS UNLESS A DIRECT-VLA SUBSET IS EXPLICITLY NAMED; NR IS COUNTED AS NOT REPORTED.

Statistic	Observed pattern
Corpus split	85 records: 30 core attack studies (23 direct VLA and 7 embodied-adjacent) plus 55 contextual references.
Direct VLA subset	23/30 core attack studies are direct VLA evidence; all percentages below use the direct subset ($n = 23$).
Entry surface flags in direct VLA	Visual/physical visual input: 14/23 (60.9%); language/instruction: 9/23 (39.1%); reasoning-trace input: 1/23 (4.3%); privacy/query output: 2/23 (8.7%); poisoned trajectory/data trigger: 1/23 (4.3%). Multi-label counts overlap.
Knowledge level	Any black-box or transfer-only condition: 13/23 (56.5%); any white/gray-box optimization condition: 14/23 (60.9%). Multi-label counts overlap when papers report multiple settings.
Access and privilege	Digital access channel: 14/23 (60.9%); physical or digital-to-physical channel: 8/23 (34.8%); supply-chain or data-contributor channel: 5/23 (21.7%).
Highest validation tier	Digital input-only: 3/23 (13.0%); simulation rollout: 10/23 (43.5%); real-sensor input: 1/23 (4.3%); real-robot actuation: 9/23 (39.1%); HIL: 0/23 (0%); human-proximate: 0/23 (0%).
Model/benchmark concentration	OpenVLA-family target appears in 10/23 (43.5%); LIBERO appears in 12/23 (52.2%); π_0/π_0 -fast appears in 6/23 (26.1%).
Reporting practices	Explicit transfer beyond one target/model/prompt: 16/23 (69.6%); limited transfer notes: 6/23 (26.1%); benign utility reported: 7/23 (30.4%); query/edit budget reported: 6/23 (26.1%).
Robot-control reporting	Action/chunk horizon reported: 3/23 (13.0%); controller frequency reported: 1/23 (4.3%); safety filter reported: 0/23 (0%); recovery policy/outcome reported: 0/23 (0%); pre/post-controller trace: 0/23 (0%).

should report not only ASR but also first affected representation, action deviation, recovery opportunity, controller intervention, and whether the safety envelope changed the outcome.

E. System, Memory, Tool, HRI, and Persistent Attacks

Embodied-adjacent evidence. Tool-agent attacks and indirect prompt injection show how untrusted context can override source boundaries and cause tool misuse or data leakage [52], [90]–[93]. In robotics, the analogous failure can become an unauthorized skill call, unsafe tool argument, or middleware command, but most evidence remains adjacent rather than directly VLA-validated. Toolformer, ReAct, Gorilla, ToolLLM, WebShop, and WebArena are useful here only as software-agent substrates; they should not be counted as VLA evidence unless robot skills or VLA-controlled actions are in the loop [94]–[99]. ROS/ROS2 and robot cybersecurity work show that middleware, topics, services, node identity, and controller interfaces can be compromised [54], [100]–[106]. HRI work shows that robots can affect human trust, authority, and physical access decisions [107]–[110].

Analogical evidence. RAG and memory poisoning show how retrieved context can be corrupted and later reused [51], [111]–[113]. For VLA systems, the unvalidated but important hypothesis is multi-session embodied memory compromise: a poisoned object memory, map, user preference, or tool log could redirect future robot behavior. This remains mostly an open VLA attack problem, not a demonstrated field-wide result.

V. CROSS-STUDY COMPARISON AND EVIDENCE GAPS

This section distills what the current evidence says across attack mechanisms. We use the core attack studies as the denominator for cross-study claims and use contextual references only to explain substrates and evaluation concepts. The key point is that attacks should be compared by mechanism, validation tier, action representation, and consequence, not by a single undifferentiated ASR.

A. What Current Evidence Tells Us

The first lesson is that VLA attacks are increasingly *representation attacks*, not only input perturbations. Visual patches matter because they can move embeddings, grounding, visual proprioception, and trajectory objectives; instruction attacks matter because text can steer plans and action tokens; backdoors matter because triggers can bind to reusable action primitives or chunks. This explains why a survey organized only by input modality misses the robotics mechanism.

The second lesson is that closed-loop evidence is still thin. Many papers show that an attack changes a rollout, but fewer show what happens between the VLA output and the physical state: controller clipping, safety filtering, latency, emergency stop, or recovery. Current evidence therefore supports claims about vulnerable VLA representations more strongly than claims about deployed robot hazard rates.

The third lesson is that the field is splitting into two complementary directions. One direction seeks stronger and more realistic attackers: black-box instruction edits, transferable patches, object-bound 3D textures, partial-observation attacks, and supply-chain triggers. The other direction seeks better evaluation: typed ASR, clean utility, transfer axes, validation tiers, and privacy metrics. A mature VLA attack literature will need both; stronger attacks without robotics-aware evaluation will not answer T-RO-relevant safety questions, and evaluation protocols without adaptive attacks will underestimate residual risk.

B. Direct VLA Evidence by Attack Mechanism

The direct matrix separates four mechanism families that are often conflated. First, instruction-control attacks operate through language while measuring robot behavior: RoboGCG asks whether adversarial text can give low-level control authority [17]; SABER asks whether bounded black-box edits can induce task failure, action inflation, or constraint violations across VLA policies [20]; Q-DIG asks whether natural, task-relevant prompt diversity can systematically expose VLA policy failures [34]; and trajectory redirection asks whether a command-preserving prompt can move the full rollout toward an attacker-specified physical outcome [31]. These are all

TABLE VI

CROSS-STUDY SYNTHESIS FOR THE 23 DIRECT VLA ATTACK RECORDS. COUNTS ARE PAPER COUNTS; ROWS MAY OVERLAP WHEN A PAPER EVALUATES MULTIPLE SETTINGS.

Attack family	Embedding/grounding	Reasoning/plan	Action token/chunk	Trajectory/controller	Dominant validation tiers
Instruction/control	1	3	2	2	Simulation rollout; one real-robot red-team study.
Visual/physical	5	1	1	6	Simulation and real-robot actuation dominate; one real-sensor-only transfer study.
Backdoor/action	1	1	4	3	Mostly simulation; three real-robot actuation reports.
Reasoning/privacy	1	3	1	2	Digital input-only and simulation; physical actuation only for patch-based reasoning hijack.
Timeline	2025: 10 direct records, mostly prompt/patch/backdoor. 2026: 13 direct records, adding black-box instruction, reasoning, privacy, 3D texture, partial-observation, and driving-trajectory evidence.				

language-entry attacks, but they differ in objective, attacker cost, and whether the attack is targeted.

Second, visual attacks split into action-loss, representation-loss, transfer, and physical-object variants. Wang et al. optimize VLA-specific action discrepancy and trajectory objectives [18]. EDPA and ADVLA move the target earlier, toward visual embeddings, projected features, attention, and vision-language alignment [27], [35]. UPA-RFAS and VLA-Hijack study transfer by exploiting shared feature-space, attention, semantic, or visual-proprioception structure rather than only a model-specific action head [28], [36]. Tex3D and partially observable patch attacks add physical realism dimensions: object-bound textures, trajectory-aware optimization, and prefix-limited attacker observation [29], [37].

Third, action/backdoor attacks are not merely training-time versions of patches. BadVLA targets trigger-conditioned action deviation [22]; GoBA targets physical-object-triggered goal behavior [23]; DropVLA targets a reusable primitive at a short reaction window [24]; AttackVLA compares multiple attack implementations and introduces long-horizon triggered action sequences [25]; FreezeVLA and SilentDrift target action freezing and chunk-level drift [21], [26]. The important comparison variable is the attacked action representation: action token, gripper primitive, static action state, full trajectory, or chunked delta-pose sequence.

Fourth, reasoning and privacy attacks expand the objective set. TRAP and altered-thought evaluations show that reasoning traces, entity references, and CoT-to-action pathways can be attack targets in reasoning-enabled VLA systems [30], [38]. ReasonBreak shows a related reasoning-trajectory vulnerability in autonomous-driving VLA systems, where closed-loop collision and safety metrics matter more than tabletop task success [39]. Membership inference and VLALeaks show that action likelihoods, action errors, temporal motion patterns, attention discrepancies, and trajectory structure can leak training membership [32], [33]. These attacks should be evaluated with confidentiality metrics, not forced into task-failure ASR.

C. What Can and Cannot Be Compared

ASR is not directly comparable across the current VLA attack literature. In one paper, ASR may mean task failure under a visual patch; in another, targeted gripper release within a reaction window; in another, action inflation; in another, trajectory redirection; and in privacy work, AUC or TPR@FPR

for membership inference. The denominator also varies: image frames, task episodes, triggered episodes, candidate trajectories, model queries, or physical trials. Validation tier varies from digital input-only to simulation rollout to real-robot actuation. Pooling these numbers would create false precision.

Comparable claims are therefore qualitative and typed. Direct evidence supports the claim that VLA policies can be attacked through vision, language, action representations, backdoors, reasoning traces, and privacy channels. It does not yet support a universal claim that any particular attack transfers across all VLA architectures, embodiments, controllers, or safety envelopes. Embodied-adjacent evidence supports risk hypotheses about source confusion, tool misuse, middleware compromise, sensor corruption, and HRI authority ambiguity. Analogical evidence supports mechanisms and evaluation requirements, not VLA vulnerability claims by itself.

D. Evidence Gaps

The strongest direct evidence is concentrated in OpenVLA and LIBERO-style manipulation. This is useful because it creates a common baseline, but it weakens claims about general VLA security. Flow/diffusion policies, bimanual systems, mobile manipulation, navigation, tactile policies, and high-level tool/skill interfaces are less consistently covered. Action representations are also unevenly studied: tokenized action heads and patch-induced trajectory deviation are common; diffusion/flow denoising paths, velocity fields, controller latency, temporal ensembling, recovery policies, and safety-envelope interactions are much less mature.

Physical validation is improving but remains narrow. Some direct VLA papers include physical robot demonstrations, but few provide enough detail on sensor stack, control frequency, calibration, lighting, object mass, latency, recovery behavior, and safety filters to make deployment claims portable. Hardware-in-the-loop is surprisingly underdeveloped as a middle tier between simulation and full physical trials.

Privacy is an emerging direct VLA attack area rather than a side issue. Membership inference papers show that VLA training data can leak through action outputs, likelihoods, attention patterns, and temporal motion signatures [32], [33]. This evidence is direct, but it supports confidentiality conclusions, not physical-safety conclusions. Future survey tables should therefore include privacy objectives rather than treating all VLA attacks as task-failure attacks.

E. Target Model and Benchmark Concentration

The target-model distribution is informative evidence. OpenVLA and OpenVLA-OFT appear frequently because they are open, strong, and easy to evaluate on LIBERO [6], [47], [85]. π_0 and π_0 -fast appear in several attacks because flow or diffusion-style action generation changes both inference speed and attack target representation [8]. SpatialVLA appears in benchmark-oriented attack work because spatial action grids and 3D spatial encoding create a different target representation from plain tokenized actions [84]. UniVLA, CronusVLA, reasoning VLAs, Alpamayo-style driving VLAs, and VLA-Adapter broaden the evidence base, but they are not yet as repeatedly tested as OpenVLA-style manipulation systems [26], [30], [36], [39].

The benchmark distribution is similarly concentrated. LIBERO is the dominant manipulation benchmark in the direct attack matrix, with RLBench, CALVIN, ManiSkill, RoboCasa, and DROID-derived data serving more often as capability or transfer context [86], [114]–[117]. This concentration is useful for comparability but risky for authority: a benchmark can overrepresent tabletop manipulation, fixed cameras, short horizons, and simplified safety envelopes. Benchmark audits and robot-data papers therefore remain relevant to attack evidence even when they are not attacks themselves [5], [87], [88], [118].

F. Reporting Coverage

The direct VLA literature reports clean utility more often for backdoors than for inference-time attacks, because a backdoor must remain hidden under benign inputs. Transfer is reported most often in visual patch and benchmark papers, but transfer axes differ: model transfer, task transfer, viewpoint transfer, suite transfer, and sim-to-real transfer are not interchangeable. Query budget appears in black-box instruction work but is absent from many visual and backdoor studies. Adaptive settings remain rare: most papers test a fixed attack against an undefended or lightly defended policy, not against a deployed prompt hierarchy, source-authentication layer, action shield, or runtime monitor. Recovery is the weakest reporting dimension. Even when physical trials are included, many papers end evaluation at task failure or target action induction rather than measuring whether the robot could stop, replan, ask for help, or recover under controller constraints.

G. Open Versus Understudied

Some topics are genuinely open: multi-session memory poisoning of VLA robots, adaptive attacks against deployed safety envelopes, HRI attacks with ethically controlled human subjects, and standardized HIL protocols. Other topics have preliminary direct evidence but insufficient breadth: transferable visual patches, black-box instruction attacks, action-level backdoors, reasoning-chain attacks, and VLA privacy attacks. These should be described as early but real VLA evidence, not as unexplored.

VI. STANDARDIZED EVALUATION PROTOCOL

VLA attack evaluation should answer four questions: what representation changed, what embodied consequence followed, what evidence level supports the claim, and whether the result survives benign utility, transfer, and recovery checks. Table VII gives a minimum reporting checklist.

A. Typed Metrics

Untyped ASR is insufficient. We therefore compute attack rates from the typed success definition in Section III-B. Let $\kappa_O(\tau^\delta)$ indicate that adversarial rollout τ^δ satisfies objective type O , let $\sigma_C(\tau^\delta)$ indicate violation of consequence class C , and let $\rho_H(\tau^\delta)$ indicate that the attack effect persists after horizon H feedback steps, replans, memory updates, or sessions. For N trials,

$$\begin{aligned} \text{ASR}_O &= \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\kappa_O(\tau_i^\delta) = 1\}, \\ \text{SVR}_C &= \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\sigma_C(\tau_i^\delta) = 1\}, \\ \text{PR}_H &= \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\rho_H(\tau_i^\delta) = 1\}. \end{aligned} \quad (4)$$

Here ASR_O may refer to untargeted failure, targeted object choice, gripper primitive induction, action freezing, trajectory redirection, privacy membership inference, unauthorized skill invocation, or persistent memory compromise. SVR_C records consequence-specific violations such as unsafe action, collision/contact risk, privacy leakage, or authorization failure. PR_H is required for backdoors, memory attacks, repeated instruction edits, action-chunk drift, and attacks that exploit recovery behavior.

For robot deployments, the attack should also be evaluated conditionally on the controller and recovery system. Let $r(\tau^\delta)$ indicate that a recovery opportunity occurred, $q(\tau^\delta)$ that the controller or safety filter rejected the command, d_{pre} and d_{post} be pre- and post-controller deviations from the benign action or trajectory, and s_{min} be the minimum signed safety margin along the rollout. Useful controller-aware metrics include

$$\begin{aligned} \text{ASR}_{O|r} &= \frac{\sum_i \mathbf{1}\{\kappa_O(\tau_i^\delta) = 1 \wedge r(\tau_i^\delta) = 1\}}{\sum_i \mathbf{1}\{r(\tau_i^\delta) = 1\}}, \\ \text{CRR} &= \frac{1}{N} \sum_i \mathbf{1}\{q(\tau_i^\delta) = 1\}, \quad \Delta_{\text{ctrl}} = d_{\text{post}} - d_{\text{pre}}. \end{aligned} \quad (5)$$

Here $\text{ASR}_{O|r}$ is recovery-conditioned ASR, CRR is controller rejection rate, and Δ_{ctrl} measures whether the controller attenuates or amplifies the model-level deviation. Time-to-violation and s_{min} are especially important for human-proximate robots: two attacks with the same ASR may differ sharply if one violates the safety envelope after one control cycle and another remains far from contact.

TABLE VII

MINIMUM REPORTING CHECKLIST FOR VLA ATTACK PAPERS. THE CHECKLIST IS ATTACK-CENTERED: IT RECORDS WHAT WAS ATTACKED, HOW THE ATTACKER OPERATED, HOW SUCCESS WAS MEASURED, AND WHAT EVIDENCE SUPPORTS PHYSICAL OR EMBODIED RELEVANCE.

Reporting item	What to specify	Why it changes interpretation	Examples of inadequate reporting
Threat model	Knowledge level, access channel, privilege level, timing, query budget, physical placement, tool privileges, training or inference access	Makes ASR comparable across white-box, black-box, physical, human-mediated, and supply-chain settings	Reporting ASR without query budget, access channel, or privilege boundary.
Evidence stratum	Direct VLA, embodied-adjacent, or analogical evidence	Prevents adjacent LLM-agent, sensor, or robot-security results from being overclaimed as VLA-validated attacks	Calling a VLM jailbreak or ROS exploit a VLA attack without testing a VLA policy.
Targeted component	Perception, grounding, planner, action decoder, skill API, controller, memory, simulator, human interface	Identifies the intermediate representation or decision changed by the attack	Calling all failures “jailbreaks” without locating the attacked stage.
Attack objective	Untargeted failure, targeted action, unsafe action, freezing, goal hijack, privacy leak, unauthorized tool use, persistence	Separates task degradation from security-relevant consequences	Treating any lower task success as attack success.
ASR denominator	Frame, prompt, episode, triggered episode, trajectory, query, physical trial, or membership candidate	Determines whether success rates can be compared or aggregated	Pooling gripper-release ASR with task-failure ASR or MIA AUC.
Validation tier	Digital input-only, real-sensor input, simulation rollout, HIL, real-robot actuation, human-proximate	Prevents simulator, sensor-only, or image-level results from being overclaimed	Describing a photographed patch as real-robot actuation.
Transferability	Across prompts, models, action representations, embodiments, sensors, environments, and controllers	Determines whether the attack is substrate-specific or system-level	Reporting one model and one scene as general VLA risk.
Stealth and cost	Perturbation visibility, physical plausibility, attack duration, number of interventions, compute and queries	Connects attack success to deployability and detectability	Ignoring that the attack requires conspicuous or repeated intervention.
Persistence and recovery	Whether the effect survives new observations, correction, memory reset, user intervention, or safe stop	Captures closed-loop and multi-session attacks	Ending evaluation at the first wrong action.
Robot-control trace	Action representation, chunk horizon, controller frequency, safety filter, recovery policy, pre/post-controller traces	Connects model-level attack success to robotics evidence	Reporting only decoded actions when the controller clips or rejects them.
Consequence severity	Safe stop, task failure, wrong-object action, safety-envelope violation, privacy/authorization violation, human-proximate harm	Distinguishes robustness failure from security impact	Collapsing all failed episodes into a single ASR.
Defense awareness	Whether attacks are adaptive to filters, prompt hierarchy, tool permissions, shields, monitors, or human approval	Prevents evaluation from measuring only obsolete non-adaptive attacks	Evaluating a defense only against attacks designed before the defense.

B. Action-Representation Reporting

Every direct VLA attack should name the action representation under test. For autoregressive action-token policies, report token probabilities or selected action bins, first changed token, token-to-control mapping, and whether early token errors propagate. For parallel action chunks, report chunk length, temporal ensemble behavior, re-planning frequency, intra-chunk open-loop duration, and whether the attack affects one step or the whole chunk. For waypoint or trajectory policies, report maximum and final end-effector/base displacement, curvature, unsafe workspace entry, and recovery latency. For diffusion or flow policies, report whether the target is the conditional representation, denoising noise, velocity field, trajectory distribution, sampler, or final controller command.

The propagation mechanism differs by representation. In autoregressive action-token policies, an early token error can be amplified by the detokenizer and by subsequent conditional token generation; the relevant variables are action frequency, token bin width, and whether gripper and pose tokens are coupled. In parallel action-chunk policies, the attack may corrupt a multi-step command before new observations arrive; chunk horizon, temporal ensemble, and replan rate determine the open-loop exposure. In waypoint and trajectory policies, the attack is geometric: displacement, curvature, workspace

boundary crossing, and tracking-controller error are more informative than token accuracy. In diffusion or flow policies, the target may be the conditional representation, denoising path, velocity field, sampler, or final trajectory distribution; small distributional changes can be smoothed by sampling but can also create plausible near-boundary trajectories. In skill/API policies, the safety problem is often authorization and precondition bypass: selecting a valid but unauthorized skill can be more consequential than perturbing a low-level motor command.

The physical controller must be part of the evaluation. A high-level action may be rejected, clipped, smoothed, delayed, or transformed by a low-level controller, safety filter, control barrier function, recovery policy, or emergency stop. A paper should therefore report both pre-controller and post-controller traces when possible. If only simulator actions are reported, claims about collision/contact risk or human safety should be marked as unvalidated physical claims.

C. Evidence-Family Protocols

Different attack families need different minimum evidence. For *instruction-control attacks*, a study should report the original instruction, sanitized perturbation class, query budget, edit distance or semantic-preservation score, whether the benign

task succeeds, and the first rollout step at which the action diverges. RoboGCG-style control-authority claims require a different denominator from SABER-style task success reduction, Q-DIG-style failure discovery, or trajectory-redirection claims [17], [20], [31], [34]. The recommended reporting unit is therefore a tuple: instruction pair, target policy, rollout seed, attack budget, action divergence, consequence, and recovery outcome.

For *visual and physical attacks*, a study should report whether the perturbation is digital noise, localized patch, printed patch, object texture, scene object, or sensor-path interference. It should also report patch area, placement distribution, view distribution, lighting, camera resolution, preprocessing, and whether optimization used action loss, embedding loss, attention loss, semantic misalignment, or trajectory loss. This distinction separates action-space patch attacks from EDPA/ADVLA-style feature attacks, UPA-RFAS/VLA-Hijack-style transfer attacks, Tex3D-style object-surface attacks, and partially observable patch attacks [18], [27]–[29], [35]–[37].

For *backdoor and supply-chain attacks*, a study should report the poisoned data fraction, whether poisoning is sample-, episode-, or trajectory-level, whether the trigger is visual, textual, state-based, object-based, or multimodal, and whether the attack survives downstream fine-tuning or data filtering. Clean success rate is mandatory because a backdoor with poor clean utility is easier to detect and less comparable to realistic supply-chain risk. Triggered metrics should separate failure, goal completion, primitive induction, action freezing, and chunk drift [21]–[26].

For *reasoning and privacy attacks*, the metric is not ordinary task ASR. Reasoning attacks should record the attacked reasoning object, such as natural-language CoT, object entity references, bounding-box traces, subgoal plans, or driving rationales, and then measure whether the changed reasoning causally changes action [30], [38], [39]. Privacy attacks should report the membership unit, access regime, candidate distribution, and confidence metric, because sample-level and trajectory-level membership inference are not comparable to physical task failure [32], [33].

D. Representation Baselines

VLA attacks often inherit representation assumptions from VLMs, action policies, and adversarial ML. Those baselines should be cited and coded as substrate or analogical evidence rather than counted as direct VLA attack evidence. Vision-language backbones such as CLIP, Flamingo, BLIP-2, LLaVA, MiniGPT-4, and InstructBLIP help explain why embedding, attention, and cross-modal alignment are plausible VLA targets [119]–[124]. Query-efficient and transfer-oriented adversarial methods help design black-box evaluation, but they are still method baselines unless a VLA rollout is attacked [125]–[133]. This separation lets a review include necessary methodological context without inflating the amount of direct VLA evidence.

E. Validation Tiers

We use six validation tiers. *Digital input-only* tests are useful for isolating mechanisms such as attention, likelihood, embeddings, prompt transfer, or membership features. *Real-sensor input* means the perturbation is captured by a real camera, microphone, LiDAR, or robot sensor but does not necessarily drive actuation. *Simulation rollout* measures closed-loop behavior in a simulator. *Hardware-in-the-loop* exposes real sensors, controller timing, middleware, or safety filters while limiting hazard. *Real-robot actuation* means the robot physically executes actions under the attack. *Human-proximate validation* is reserved for studies that place people or human-subject interaction inside the evaluated safety boundary. Hybrid attacks should name both the delivery channel and the consequence tier, for example digital prompt with real-robot actuation, or physical patch with simulation-only consequence.

F. Benchmark Use

Benchmarks must be matched to claims. LIBERO, CALVIN, RL Bench, ManiSkill, RoboCasa, and DROID-derived settings can support manipulation and rollout comparisons, but they do not by themselves establish physical safety. AgentDojo, ToolEmu, InjecAgent, and RAG benchmarks support source-boundary and tool-risk analogies, not robot consequences. Embodied jailbreak benchmarks support action-boundary evaluation for planners and LLM-controlled robots, but direct VLA claims require VLA policies or VLA-enabled robot systems. A credible attack benchmark needs paired benign/adversarial tasks, evidence strata, knowledge/access/privilege coding, ASR definition, severity labels, transfer tests, validation tier, and recovery metrics.

G. Adaptive and Responsible Evaluation

Guardrails change the attack objective. Prompt hierarchy turns injection into source-boundary confusion; memory hygiene turns poisoning into persistence; tool permissions turn misuse into argument selection or authorization-boundary crossing; safety filters turn direct collision objectives into near-boundary, semantic misuse, or recovery-exhaustion objectives. Attack papers should state whether the attacker knows the guardrail, receives rejection feedback, adapts to it, or acts only through the physical environment.

Because VLA attacks may transfer to real robots, papers should publish sanitized prompts, aggregate statistics, threat models, and defense-relevant analysis while withholding reusable high-risk payloads or hardware exploit recipes when needed. Responsible disclosure is part of evaluation quality, not an afterthought.

VII. OPEN PROBLEMS

A. From Direct Evidence to General Evidence

Direct VLA attack evidence is no longer absent, but it is not yet broad enough for field-wide conclusions. The next step is replication across action-token, chunked, diffusion, flow, waypoint, skill-based, and hierarchical VLA systems; across

TABLE VIII

VALIDITY OF COMMON TESTBEDS FOR VLA ATTACK CLAIMS. A BENCHMARK SUPPORTS ONLY THE EVIDENCE LEVEL IT ACTUALLY MEASURES.

Benchmark/framework	Type	What it supports	Invalid claim if used alone	Attack overlay needed
AI2-THOR / ALFRED / Capability Habitat [134]–[136]		Grounding and household/navigation planning	Physical safety or controller-level attack success	Adversarial goals, source conflicts, recovery labels.
RLBench / CALVIN / Capability LIBERO / ManiSkill [47], [114]–[116]		Manipulation trajectories and language-conditioned tasks	Security claims from task success alone	Targeted ASR, trajectory deviation, paired benign/adversarial tasks.
RoboCasa / DROID-derived data [86], [117]	Data/testbed	Scene diversity and data provenance questions	Poisoning or sim-to-real claims without attack labels	Provenance checks, poisoned-demo splits, validation tiers.
ToolEmu / AgentDojo / In-jecAgent [52], [92], [93]	Red-team	Tool misuse and indirect prompt injection	Direct VLA or physical consequence claims	Robot skill permissions, physical-action proxy, latency.
SafeAgentBench / EARBench [82], [83]	Safety	Hazard/risk-aware planning	Low-level VLA action or hardware safety	Action traces, severity labels, adaptive attacks.
BadRobot / RoboPAIR / RoboJailBench [40], [41], [43]	Attack/red-team	Embodied jailbreak goals and security-utility pairs	Cross-platform VLA generality without more models	Paired benign goals, physical constraints, model transfer.
RoboGCG / VLA-Fool / SABER [17], [19], [20]	Direct VLA attack	Prompt, multimodal, and black-box instruction objectives	Mature field-wide evidence; many are preprints	Cross-policy transfer, query budgets, physical rollouts.
Patch/physical VLA attacks [18], [27]–[29]	Direct VLA attack	Visual embeddings, grounding, textures, trajectories	Universal physical safety claims from narrow scenes	Viewpoint, object, controller, and sim-to-real transfer.
FreezeVLA / DropVLA / AttackVLA [21], [24]–[26]	Direct VLA attack	Action freezing, primitives, chunks, backdoors	General action-security evidence across all VLA representations	Token/chunk/flow transfer and recovery metrics.
CHAI / ANNIE / Eva-VLA / Phantom Menace [42], [44]–[46]	Adjacent physical/sensor attack	Scene text, trajectory, physical variation, sensor path	Direct VLA attack claims without VLA policy validation	Threat checklist, recovery, cross-platform validation.
Benchmark audits [118]	Validity diagnostic	Shortcut, overfit, and statistical fragility	Attack benchmark evidence	Security splits and adversarial diagnostics.

manipulation, navigation, mobile manipulation, and human-proximate settings; and across controllers and safety filters. A result on OpenVLA/LIBERO should be reported as important direct evidence, not as proof of universal VLA vulnerability.

B. Physical Recovery and HIL Protocols

The missing middle tier is hardware-in-the-loop. Simulation gives scale, and physical robots give realism, but HIL can expose real sensors, middleware timing, safety filters, and controller latency without requiring unsafe full deployment. Standard HIL protocols should record when an attack first changes representation, whether a controller clips or rejects it, whether the robot recovers after new observations, and whether human intervention changes the outcome.

C. Action-Representation Transfer

The field needs attacks and defenses that test transfer across action representations, not only across model names. A prompt attack may transfer across language-conditioned token policies but fail against a flow policy; a patch may transfer across visual encoders but not across camera placements; a chunk-drift backdoor may be irrelevant to a one-step replanner. Evaluation suites should vary tokenizer, chunk length, temporal ensemble, diffusion/flow sampler, waypoint controller, and low-level safety envelope.

D. Memory, Tool, Middleware, and HRI

Direct VLA evidence is still thin for memory poisoning, tool/API misuse, middleware compromise, and socially situated attackers. Adjacent evidence from RAG, software agents, ROS/ROS2, and HRI shows credible mechanisms [51], [52], [54], [107], but VLA attack studies need to measure robot consequences: unauthorized skill use, corrupted maps, persistent user preference changes, unsafe tool arguments, privacy leakage, and recovery after correction.

E. Privacy as a First-Class VLA Attack Objective

Membership inference work shows that VLA actions and trajectories can leak training membership [32], [33]. Future privacy evaluation should extend beyond membership to property inference, dataset provenance, user routine leakage, private-scene memorization, and cross-session log leakage. These are direct VLA security questions even when no robot collision or task failure occurs.

F. Adaptive Attacks and Responsible Artifacts

First-generation defenses will use prompt hierarchy, source labels, retrieval filters, tool permissions, runtime monitors, action shields, control barrier functions, and human approvals. Attack research must therefore include adaptive settings: what the attacker knows, what feedback is available, whether queries are budgeted, and whether benign utility is preserved. At the same time, VLA attack artifacts can be unusually transferable to physical systems. Benchmarks should separate public scoring code and sanitized examples from high-risk payloads, physical exploit details, or deployment-specific trigger recipes.

G. Limitations of This Review

This review freezes the evidence database at the 16 June 2026 search date, so later arXiv releases, renamed project pages, and venue updates are outside scope. Google Scholar and Semantic Scholar ranking can bias discovery toward highly visible papers, while targeted-name searches can miss attacks that use different terminology. We searched English-language sources and may miss non-English reports. Several direct VLA attacks are recent preprints or workshop papers; their claims may change after peer review, and project-page or repository information is weaker evidence than archival text.

The complete database is auditable, but it records what papers report, not independent reproduction of their experiments. The independent second audit covers all 23 direct VLA records

and principal fields, plus a stratified sample of non-direct records, but it does not re-run attacks, inspect all source code, or verify every ASR number. Coding remains partly judgment-based, especially for multi-label target representations, transfer reporting, and validation flags. Finally, the descriptive statistics are counts and proportions over a small database, not a meta-analysis. We do not pool ASR values across attack objectives, denominators, validation tiers, or robot platforms, and multi-label percentages can change noticeably when a single new direct VLA paper is added.

VIII. CONCLUSION

This review reframes VLA attacks as a closed-loop robotics problem. The field now has direct evidence for adversarial patches, black-box instruction attacks, action freezing, action-level and long-horizon backdoors, transferable and 3D object attacks, reasoning-chain hijacking, trajectory redirection, and membership inference. That evidence is important, but it is not yet uniformly authoritative: it is recent, often preprint-heavy, concentrated around OpenVLA and LIBERO, and uneven in physical recovery, HIL, controller reporting, and adaptive evaluation.

The main conclusion is therefore evidence-aware but mechanism-driven. Direct VLA studies support claims about demonstrated VLA attack surfaces. Embodied-adjacent studies motivate hypotheses about planners, tools, sensors, middleware, and HRI that still need VLA validation. Contextual studies explain mechanisms and evaluation design, not VLA vulnerability by themselves. A shared typed definition of attack success is necessary for this field: task failure, targeted action, safety violation, authorization failure, privacy leakage, persistence, and recovery failure are different predicates and should not be collapsed into one ASR. Progress now depends on cross-representation transfer studies, physical and HIL protocols, privacy-aware evaluation, adaptive attacks against deployed guardrails, and responsible artifact release suited to embodied systems.

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